VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE

Department of Computer Science and Artificial Intelligence

B.Tech. Third Year

PROJECT SYNOPSIS

Ethical and Responsible AI

Project Title

Evaluating the Impact of Fairness Mitigation on Explanation Consistency in Machine Learning-Based Credit Scoring

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Submitted as part of the course requirements for

1. Project Title

Evaluating the Impact of Fairness Mitigation on Explanation Consistency in Machine Learning-Based Credit Scoring

2. Abstract

Credit-scoring models are increasingly used to support financial decision-making, but predictive performance alone does not establish whether such models are responsible. Existing research has separately investigated predictive performance, algorithmic fairness and explainability, while recent literature highlights the need for more integrated empirical evaluation of these dimensions. This project investigates how fairness-mitigation strategies affect both the predictive performance and local explanation behaviour of machine-learning-based credit-scoring models. Logistic Regression, Random Forest and XGBoost will be evaluated on publicly available credit datasets using predictive-performance and group-fairness metrics. Representative pre-processing, in-processing and post-processing fairness-mitigation techniques will then be applied and their effects on model performance and fairness will be measured. SHAP will be used to analyse local model explanations, with matched applicants used to investigate explanation consistency before and after fairness mitigation. The study will further analyse the trade-offs among predictive performance, fairness and explanation consistency across model–mitigation configurations. A Streamlit-based prototype will demonstrate the proposed audit methodology as a decision-support tool for credit-model analysis rather than an automated loan-approval system.

3. Introduction

Machine learning is increasingly used in financial applications such as credit scoring and loan-risk assessment, where models analyse applicant information to predict whether an applicant represents a low or high credit risk.

However, high predictive accuracy alone does not guarantee that a model is responsible. Machine learning models can learn unwanted patterns from historical data and may produce different outcomes for different demographic groups, raising concerns of fairness and discrimination. A related issue is explainability: applicants and financial institutions need to understand why a model produced a particular decision. Explainable AI techniques such as SHAP can identify the features that contributed to a prediction.

This project therefore focuses on two aspects of responsible AI: (1) the fairness of credit-risk predictions, and (2) the consistency of explanations for similar applicants.

Recent literature indicates that predictive performance, fairness and explainability in AI-based credit scoring are frequently evaluated as separate dimensions. While existing studies have investigated fairness–performance trade-offs and explainability of credit models independently, the interaction between fairness mitigation and model explanations requires further empirical investigation. In particular, understanding whether interventions that improve group-level fairness also alter the consistency of local explanations can provide additional evidence for responsible deployment of AI-based credit-scoring systems.

4. Problem Statement

Existing research has extensively investigated predictive performance, group fairness and explainability in credit scoring, but these dimensions are not always evaluated together within a common experimental framework. Furthermore, while fairness-mitigation methods can alter model behaviour, their effect on local explanations and explanation consistency for comparable applicants is less consistently evaluated across different model families and mitigation strategies. This project therefore investigates the relationship between fairness mitigation, predictive performance and explanation consistency in machine-learning-based credit scoring.

5. Motivation and Need

A loan decision can significantly affect an individual's financial opportunities, so an AI-based credit assessment system must be more than accurate — it should also be fair, explainable, transparent, consistent and responsible. This project is motivated by the growing use of AI in financial decision-making and This motivates an empirical investigation into whether conventional group-level fairness metrics provide sufficient information about changes in local model explanations following fairness mitigation.

6. Objectives

- To benchmark Logistic Regression, Random Forest and XGBoost for credit-risk prediction using publicly available credit datasets.
- To evaluate the predictive performance and group fairness of the selected credit-scoring models using appropriate performance and fairness metrics.
- To implement representative pre-processing, in-processing and post-processing fairness-mitigation strategies and quantify their impact on model performance and fairness.
- To use SHAP to analyse local explanations generated by the selected credit-scoring models before and after fairness mitigation.
- To investigate explanation consistency among comparable/matched applicants and quantify changes in explanation similarity across different fairness-mitigation strategies.
- To analyse the trade-offs among predictive performance, fairness and explanation consistency and demonstrate the resulting audit methodology through an interactive prototype.

7. Scope of the Project

Included

Public UCI credit-risk datasets

Primary evaluation on Taiwan Credit Card Default dataset

Secondary validation on German Credit dataset

Logistic Regression, Random Forest and XGBoost

Predictive-performance evaluation

Group-fairness evaluation

Representative fairness-mitigation methods

SHAP-based local explanation analysis

Matched-applicant explanation-consistency analysis

Statistical comparison of explanation changes

Performance–fairness–explanation trade-off analysis

Interactive model-auditing prototype

Not Included

Real bank loan decisions or approvals

Real customer financial data

Direct integration with banking systems

Autonomous loan approval or rejection

The system is a research prototype and is not intended as a real financial decision-making system.

8. Literature Survey / Existing Systems

Sr. No.	Papers	Author / Year	Method / System	Key Finding	Limitation / Gap
1	Fairness in Credit Scoring: Assessment, Implementation and Profit Implications	Kozodoi, Jacob & Lessmann, **2022**	Evaluates multiple fairness criteria and fairness-processing techniques across **7 datasets**; studies fairness–profit trade-offs.	Fairness can be incorporated into credit scoring, but different fairness criteria can give different conclusions; fairness improvements can involve performance/profit trade-offs.	Focuses mainly on **outcome fairness and profit** rather than whether explanations remain consistent for similar applicants.
2	Explainable AI for Credit Assessment in Banks	Bussmann et al., **2022**	Evaluates multiple fairness criteria and fairness-processing techniques across **7 datasets**; studies fairness–profit trade-offs.	Demonstrates how SHAP can make high-performing credit-risk models more interpretable and suitable for financial decision-making.	Primarily addresses **explainability**, rather than jointly evaluating fairness and explanation consistency across demographic groups.
3	Interpretable Machine Learning for Imbalanced Credit Scoring Datasets	2023	Studies LIME and SHAP on highly imbalanced mortgage/credit datasets and examines explanation stability as imbalance changes.	SHAP and LIME explanations become **less stable as class imbalance increases**, showing that explanation reliability itself needs evaluation.	Focuses on **class imbalance and explanation stability**, not demographic fairness or matched-applicant explanation consistency.
4	How Fair is Machine Learning in Credit Lending?	Babaei & Giudici, **2024**	Empirically evaluates fairness of ML-based credit-lending decisions.	Highlights the need to assess fairness when ML is used for accepting/rejecting credit applications.	Concentrates on **fairness of lending outcomes**, leaving explanation-level consistency largely unexplored.

5	Measuring Fairness in Credit Ratings	Chen, Giudici, Liu & Raffinetti,	Proposes fairness metrics based on **Shapley–Lorenz values** and evaluates them on SME credit data	Demonstrates that fairness can vary between population groups and proposes a fairness measurement approach specifically for AI credit scoring.	Uses Shapley-based fairness measurement, but does not focus on whether **individual explanations for similar applicants** remain consistent.
6	The Road to Explainability is Paved with Bias: Measuring the Fairness of Explanations	Balagopalan et al., **2022**	Audits explanation quality across protected subgroups using multiple explainability methods and datasets, including finance.	Finds that explanation fidelity can differ significantly between demographic groups; explanation quality itself can therefore be **unfair**.	Not specifically designed for credit-risk assessment and does not combine its explanation-fairness analysis with the credit-risk fairness-mitigation pipeline proposed in your project.
7	Performance, Fairness, and Explainability in AI-Based Credit Scoring: A Systematic Literature Review	2026	Systematically reviews **43 peer-reviewed studies (2020–2025)** covering performance, fairness and explainability in AI credit scoring.	Finds that performance, fairness and explainability are often studied **in isolation**, while their interaction and joint treatment remain underexplored	This directly motivates your project: a unified evaluation of **performance + fairness + explanation consistency**.

9. Proposed Methodology

Input / Data Source: UCI German Credit dataset (public credit-risk data).

Data Pre-processing: missing-value handling, categorical encoding, numerical feature processing, feature selection, train–test splitting.

AI / ML Technique: Logistic Regression, Random Forest and XGBoost for prediction; SHAP for explainability; Fairlearn-based techniques for fairness mitigation.

System / Application Development: interactive dashboard (Streamlit) integrating predictions, SHAP explanations and fairness indicators.

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC for performance; Demographic Parity, Equal Opportunity, Equalized Odds, Disparate Impact for fairness; explanation-similarity measures for matched applicants.

Deployment / Demonstration: a working Streamlit dashboard demonstrating applicant input, prediction, risk score, SHAP explanation, fairness metrics and explanation-consistency results.

9.1. Proposed System Architecture / Flow

Public Credit Dataset (UCI German Credit)

↓

Data Preprocessing

↓

Feature Engineering

↓

Train / Test Split

↓

ML Models: Logistic Regression | Random Forest | XGBoost

↓
Prediction & Performance Evaluation
↓
Fairness Evaluation
↓
Fairness Mitigation
↓
SHAP Explainability
↓
Matched Applicant Analysis
↓
Explanation Consistency Evaluation
↓
Interactive Dashboard

10. Ethical and Responsible AI Considerations

Ethical / Responsible AI Aspect	Risk in Our Project	Proposed Mitigation
Fairness / Bias	Models trained on historical credit data (e.g., the UCI German Credit dataset) may learn patterns that disadvantage certain demographic groups, leading to unequal predictions.	Evaluate Demographic Parity, Equal Opportunity, Equalized Odds and Disparate Impact for all models; apply Fairlearn-based mitigation techniques and compare fairness before/after mitigation.
Privacy & Data Protection	Fairness analysis requires access to demographic attributes, which are sensitive.	Use only the public, anonymized UCI German Credit dataset; no real customer or bank data is collected; sensitive attributes are used strictly for fairness auditing, not stored or exposed in the dashboard.
Transparency / Explainability	Random Forest and XGBoost behave as black-box models, making it hard for applicants or reviewers to understand individual decisions.	Use SHAP to generate feature-level explanations for every prediction and display them on the dashboard alongside the risk score.
Accountability	Ambiguity about who is responsible if a prediction or explanation is wrong or inconsistent.	Clearly scope the system as a research prototype (Section 8: Not Included — real loan decisions); document model versions, preprocessing steps and evaluation results for traceability.
Safety / Reliability	Small dataset size and class imbalance could make models unstable or produce inconsistent explanations for similar applicants.	Use fixed random seeds, documented preprocessing/model configuration, and matched-applicant testing to check the stability of predictions and explanations.
Human Oversight	Risk of the dashboard being mistaken for an automated approval system.	Position the dashboard explicitly as a decision-support and analysis tool for human reviewers, not an automated loan-approval system.

10.1. ART of AI

Accountability: The project team is answerable for the correctness of the models, the validity of the fairness and explanation-consistency analysis, and for clearly communicating that this is a research prototype and not a real credit-decision system.

Responsibility: The team is responsible for dataset selection, preprocessing choices, model development, fairness evaluation/mitigation, and for monitoring and documenting the trade-offs between accuracy and fairness.

Transparency: Users of the dashboard will be able to see the predicted risk, the SHAP-based reasons behind each prediction, the fairness metrics across demographic groups, and the known limitations of the system.

11. Expected Outcomes

A working credit-risk prediction system trained and compared across multiple ML models.

Quantitative performance evaluation (Accuracy, Precision, Recall, F1-score, ROC-AUC).

Fairness evaluation across demographic groups and fairness-mitigation experiments.

SHAP-based individual explanations and matched-applicant explanation comparisons.

An explanation-consistency analysis identifying when similar applicants receive dissimilar explanations.

An interactive dashboard visualising predictions, explanations and fairness indicators.

Experimental findings suitable for a research paper on fairness and explanation consistency in credit-risk AI.

12. Tools and Technologies

Category	Tool / Technology	Purpose
Programming Language	Python	Core development language
Data Processing	Pandas, NumPy	Data cleaning, transformation, numerical computation
Machine Learning	Scikit-learn, XGBoost	Model training: Logistic Regression, Random Forest, XGBoost
Explainable AI	SHAP	Feature-level explanation of individual predictions
Fairness	Fairlearn; custom fairness-metric functions	Fairness evaluation and mitigation
Visualization	Matplotlib, Seaborn, Plotly	Charts and exploratory/result visualization
Dashboard	Streamlit	Interactive dashboard for predictions, explanations and fairness results
Development Environment	VS Code, Jupyter Notebook / Google Colab	Development and experimentation
Dataset	UCI German Credit Dataset	Public credit-risk dataset for training and evaluation

13. Project Plan / Timeline

Activity	Week 1	Week 2	Week 3	Week 4+
Literature Survey & Problem Definition	✓			
Dataset Collection & Preprocessing	✓	✓		
Model Development (LR, RF, XGBoost) & Evaluation		✓	✓	
Fairness Evaluation & Mitigation			✓	✓

SHAP Explainability & Matched-Applicant Analysis			✓	✓
Dashboard Development, Testing & Documentation				✓

14. Deliverables

Working prototype: fairness-aware, explainable credit-risk prediction system

Source code and documentation

Model performance, fairness and explanation-consistency evaluation results

Ethical-risk / responsible-AI analysis (Section: Ethical and Responsible AI Considerations)

Interactive dashboard, project presentation and demonstration

15. References

[1] N. Kozodoi, J. Jacob, and S. Lessmann, "Fairness in credit scoring: Assessment, implementation and profit implications," *European Journal of Operational Research*, vol. 297, no. 3, pp. 1083–1094, 2022. DOI: 10.1016/j.ejor.2021.06.023.

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[4] S. Das, R. Stanton, and N. Wallace, "Algorithmic Fairness," *Annual Review of Financial Economics*, vol. 15, pp. 565–593, 2023. DOI: 10.1146/annurev-financial-110921-125930.

[5] G. Babaei and P. Giudici, "How fair is machine learning in credit lending?" *Quality and Reliability Engineering International*, vol. 40, pp. 3452–3464, 2024. DOI: 10.1002/qre.3579.

[6] Y. Chen, P. Giudici, K. Liu, and E. Raffinetti, "Measuring fairness in credit ratings," *Expert Systems with Applications*, vol. 258, Art. 125184, 2024. DOI: 10.1016/j.eswa.2024.125184.

[7] A. Balagopalan, H. Zhang, K. Hamidieh, T. Hartvigsen, F. Rudzicz, and M. Ghassemi, "The Road to Explainability is Paved with Bias: Measuring the Fairness of Explanations," 2022.

[8] "Performance, Fairness, and Explainability in AI-Based Credit Scoring: A Systematic Literature Review," *Journal of Risk and Financial Management*, 2026.

16. Student Declaration

We declare that the proposed project synopsis is prepared by us for the course Ethical and Responsible AI. The information and references used will be appropriately acknowledged, and the project will be developed with due consideration to fairness, privacy, transparency, accountability, safety, and human oversight.

Student 1	Student 2	Student 3	Student 4
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17. Guide / Faculty Remarks